

Business Problem Statement

A retail company has collected customer shopping data containing information about customer demographics, purchased products, spending, discounts, subscriptions, shipping preferences, purchase frequency, and product ratings.

The company wants to understand how different types of customers behave and which products, categories, and customer segments contribute more to overall sales. The management also wants to identify purchasing patterns that can help improve customer engagement and make better sales and marketing decisions.

The analysis should answer the following key business question:

“How can customer purchase data be analyzed to understand customer preferences, identify high-value customer segments, and improve sales and customer retention strategies?”

Deliverables

- 1. Data Preparation & Analysis (Python):** Clean and transform the raw customer data and prepare it for analysis.
- 2. SQL Analysis (PostgreSQL):** Analyze customer spending, product ratings, discounts, subscriptions, repeat purchases, and revenue using SQL queries.
- 3. Power BI Dashboard:** Develop an interactive dashboard to present key customer and sales metrics through visualizations and filters.
- 4. Business Insights:** Summarize important patterns and findings from the analysis to support data-driven business decisions.